Information Retrieval

Prof. Alberto Sillitti

3. Quality assessment

INNOPOLIS UNIVERSITY

Assessment

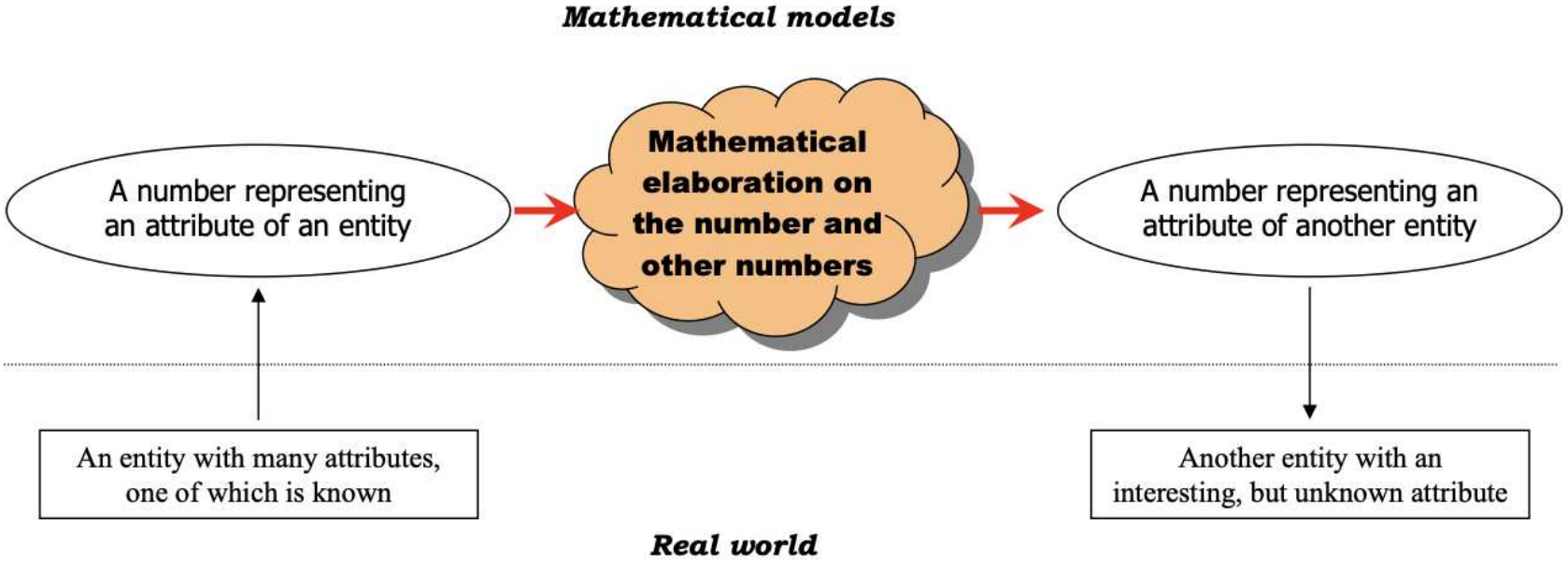
- Assessing a problem is essential
- Assess means **evaluate** a problem with **quantitative** and **qualitative** analyses

- Measurement's **purposes**:
  - planning
  - managing
  - reviewing

"If you cannot measure, you can not improve!” (Tom DeMarco)

Measurement

- Is a **process** by which we assign a **number** to something so that, given two (some) things, we can know which one is more.



Representational Theory of Measurement

1) Identify the **entities** that we want to measure
2) Identify their **measurable** attributes
3) Assign **symbols** to the attributes, moving from the real world to the mathematical one
4) Identify **relationships** among symbols
5) Transpose the discovered relationships in the world of numbers to the real world

Representational Condition

- Risks
  - assigning the **wrong number** to attributes of entities
  - transposing the relationship among entities back into the **wrong relationship** in the real world

- In the representational theory of measurement, measurement is the process of assigning symbols to attributes of entities, so that with the process of transposing entities between the real world and the world of symbols, the **meaning of the relationships is preserved**

- Bad measure: measure that does not satisfy this condition

What is the target metric of a service?

- Money
- Traffic share (yandex vs google)
- User satisfaction
- User happiness
- Logins/Subscriptions
- ...

1) Unfortunately, complex human behaviors are difficult to model (e.g., if we had this feature, we would attract $$)
2) Some targets are hard to compute online (delays, is it measurable?, ...)

So, what?

- Create a set of easily computable metrics which correlate with target ones.
  - Satisfaction (?) vs probability to find relevant doc
  - Traffic Share (delayed) vs MAU (Montly Active Users)

High-level evaluation techniques

- **Offline evaluation** — result of a new model is compared with *manually [pre]processed data*
  - ***On a bucket***: accuracy, precision, recall
  - ***By assessors***: relevance scale
    - (assessors are humans: kappa stats, weighted, majority voting, …)
- **Online evaluation** — a new model is *compared* with an old one by some *target metric* on a *subset of users*. Most widely used approach is A/B testing
- **Readings:** https://dl.acm.org/doi/abs/10.1145/3329188



What is relevance?

- Information need is encoded in a query.
- Query is used to get results.
- But do results satisfy information need?
- {0, 1} or [0..1]: {IR, REL-, REL+}

Online: A/B testing

Online testing allows to test of a *target metric*!

1. Is *relative*. Create **alternative** model
2. Formulate **target metric** (CTR, dwell-time, total revenue, ARPU)
3. Prepare **representative subset of users** (for complex setups use multinomial experiments)
4. Run controlled experiment, **aggregate statistics** on your metric
5. Run **statistical test** (Welch, Fisher, Student, …) to see that **mean shift** for a distribution is significant*
6. Make a **decision** to accept (no always ‘“better” for new features)

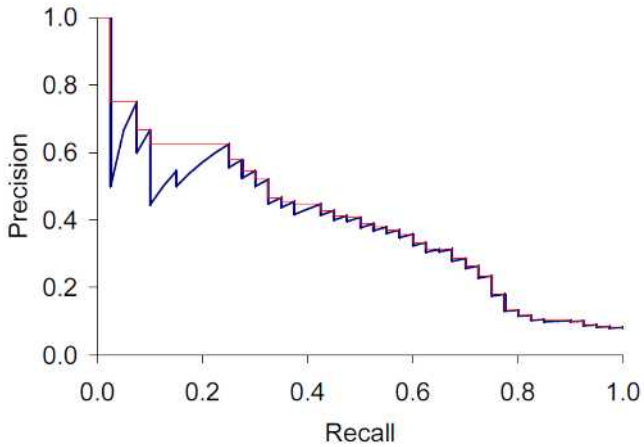
A/B testing infrastructure

- User **identification** method (logged in, cookies, fingerprints)
- **Experiment support** in the whole infrastructure (should not be as usual traffic) with flags/parameters
- Mapping **registry** (users to experiments)
- **Statistical tools** ready before you start experiment

Offline: accuracy, precision, recall, F_1

- **Accuracy** is not the case
- **Precision** is how many relevant out of retrieved
- **Recall** is how many relevant retrieved out of all relevant
- **Precision-recall curve** can be used for ranked results
- F_1 is to come up with a single number

- Users are *tolerant* to irrelevant results



Average Precision (AP)

- It summarizes the performance of a ranked retrieval system by taking the **mean of precision scores** at the ranks of all relevant documents, or the area under a Precision-Recall curve.
- It quantifies how well a model identifies **relevant** items in a ranked list, with **higher AP** indicating **better performance** in finding relevant items and placing them higher in the ranking

Mean Average Precision (mAP)

- It is calculated by taking the **average of the Average Precision (AP)** scores over **multiple queries**.
- A higher mAP indicates a better-performing system, as it signifies that relevant items are ranked higher in the results

Objective

- User should solve a problem with a service: get an answer to the question, a service or an item without leaving portal. Steps to achieve:
  - Keep users logged in
  - Evaluate user intent (surfing, buying, asking, ...)
  - Provide a quality service for each intent (first, specific search, then specific service)